Video Transcript

Video 10: Serverless Cloud Database Overview (04:00)

By now, you have a pretty good sense of databases. You've used the traditional relational database, you've used document databases, you've used key value store databases and distributed scalable databases. But you might be wondering about how this looks like in the cloud and perhaps of buzzwords such as serverless. What does that mean? Well, let's take a step back and consider where we have been and what we have done. We started out by doing a local installation in the traditional way. Navigate to a website, download a program, run it locally and configure it. And really this picture of what you see here is a conceptual one, because really what we did is we downloaded that database, installed it, ran it locally, and then built some program to access the database.

We ran a Python program that used the driver to be able to access and interact with the database. Now, recognizing all of the complexity that that entails, we took a shortcut and we used containers, and those containers could be loaded from a registry of images and then run locally on top of our Docker runtime. Now, you can imagine that you could do the same thing in the cloud. Within the cloud, you can provision a machine and on top of that add the Docker runtime, and on top of that add those containers or those images and run containers locally.

Very much the same model that we're using locally, but now in the cloud. And certainly that gives us a lot of advantages. It certainly gives us portability, but it would be nice to have more. What if the cloud provider could do those installations for you and simply offer you the service? That is, you wouldn't have to think about what operating system it was installed on. You wouldn't have to think about things such as scalability. You would simply use the database offering and let the cloud provider solve those problems. Well, that exists and it's called 'Manage Cloud Services', and it's something that might be a great resource for you to use if it fits your use case.

Now, what if we wanted more, if we were greedy and we wanted even more simplicity? So, this is what we refer to when we think about serverless. That is, this scenario in which you no longer are defining the operating system, neither the database that is being used. You're simply being offered functionality of a database through an API application programming interface, so there's no decision on whether you will use a document or a relational or a key value or other type of data store. That is an implementation detail that the cloud vendor decides on.

And all you really care about is that the information you are inputting and recalling meets a certain level of service level agreement. And if that is matched, then this is this is the performance you want. And in some scenarios this is a perfect solution. It lets you bypass everything else that we have considered. Now, in this last example, when it comes to databases, we will be looking precisely at that scenario and we will be using a cloud database, a serverless database called Firebase. And you can navigate here to that address firebase.google.com. Within it, once you access that address, you can go to the console and add a new project. Now, once we are inside of the project, we will be creating a database and then just like we did before, from our local environment, we will be accessing that database that is running in the cloud in a serverless model.